

CariSurg MedTech Pathways 2026

Week 5 Interim Submission

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Programme: CariSurg MedTech Pathways 2026

GitHub Repository

<https://github.com/baldiea75-tech/carisurg-portfolio.git>

Methodology

The Yale EMLLC Admission Prediction Triage dataset was loaded into a pandas DataFrame consisting of 55,121 emergency department encounters and 225 columns. Columns were grouped into families, including the target label, vital signs, demographics, administrative fields and chief complaint flags, to make the profiling manageable.

Structural profiling was carried out using pandas' info, describe and shape functions, alongside a manual dtype audit to check for columns stored in a type that did not match their true meaning. Missingness was measured for every column as both a raw count and a percentage, and a heatmap was produced to confirm the result visually.

Vital sign values were checked against physiologically plausible ranges. Any value outside the plausible range was treated as a data error, replaced with a missing value and imputed using the column median. Correlations between the target label and both the vital signs and the chief complaint flags were then calculated to identify which features are most associated with higher acuity.

A set of four charts was produced to visualise the target distribution, patient demographics, vital sign distributions and the most common presenting complaints. Across all charts the colour scheme was kept consistent, using purple, pink and yellow as the standard palette, with red reserved only as an accent for values that had been flagged as missing or clinically impossible.

Data Quality and Exploration Graphs

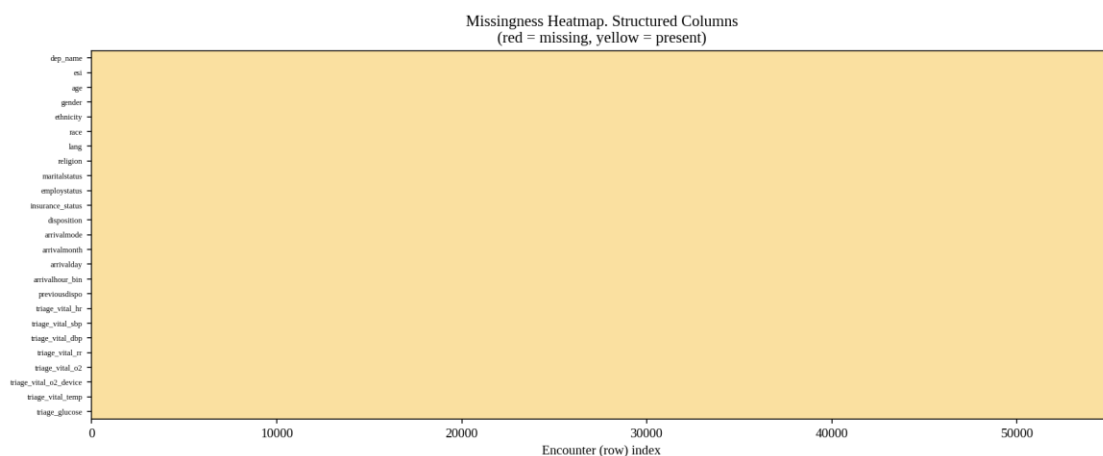


Figure 1. shows a missingness heatmap whether each structured column is present or missing across all 55,121 encounters. Yellow represents a present value and red would represent a missing value. The chart shows no red anywhere, confirming that there is zero missingness across all structured columns.

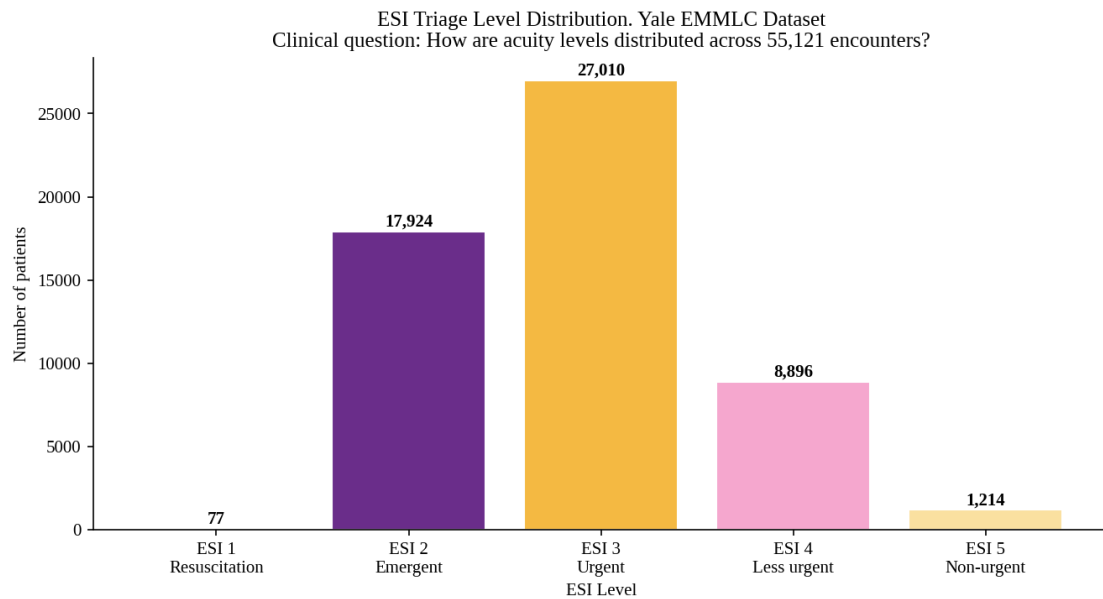


Figure 2. a ESI triage level distribution bar chart that shows how many patients fall into each of the five ESI triage levels, where level 1 is the most critical and level 5 is the least critical. Most patients fall into ESI 2 and ESI 3, while ESI 1 contains only 77 patients, showing a strong class imbalance in the most urgent category.

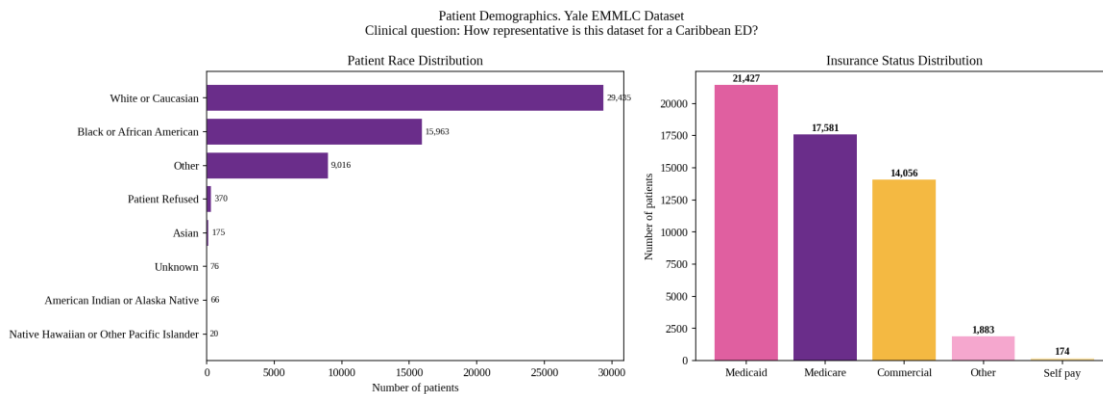


Figure 3. shows patient demographics, the left panel shows the distribution of patient race, and the right panel shows the distribution of insurance status. Most patients are recorded as White or Caucasian or Black or African American, and most visits are covered by Medicaid or Medicare, giving an overall picture of the patient population represented in this dataset.

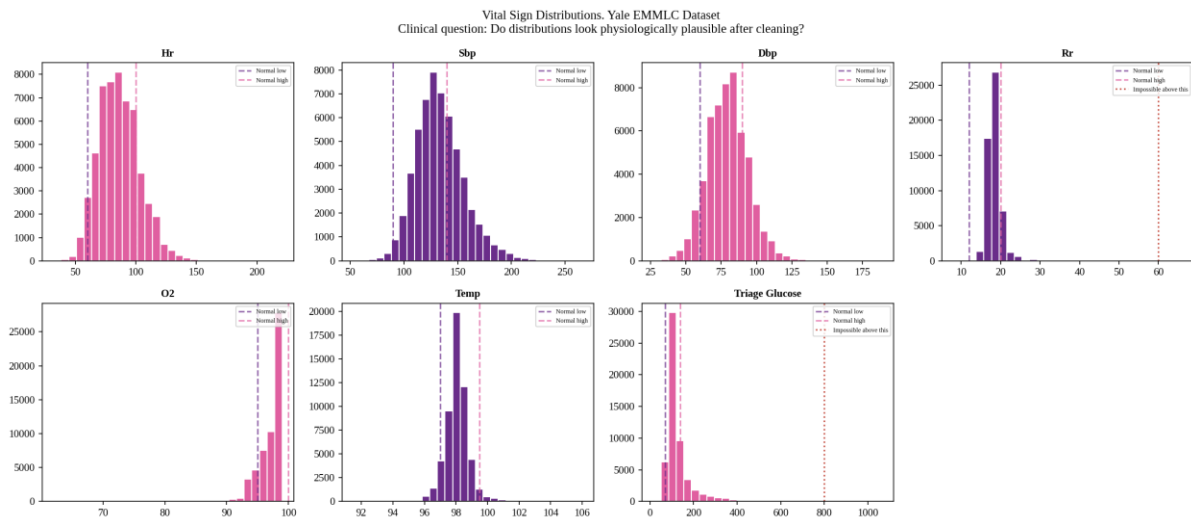


Figure 4. vital sign distributions, these histograms show the distribution of each of the seven triage vital signs after cleaning. The purple and pink dashed lines mark the normal clinical range for each vital sign, and the red dotted lines on respiratory rate and blood glucose mark the clinically impossible cutoff used to flag and correct erroneous values. All distributions fall within a physiologically plausible shape after cleaning.

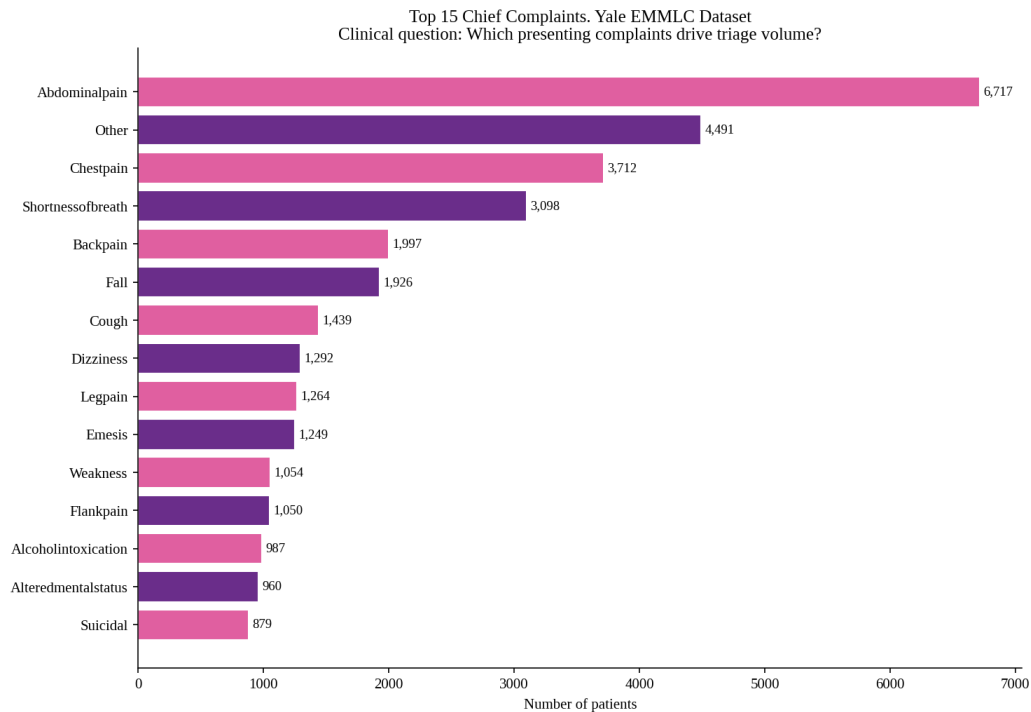


Figure 5. shows top chief complaints, a horizontal bar chart shows the fifteen most common presenting complaints recorded at triage. Abdominal pain is the most frequent complaint, followed by chest pain and shortness of breath, both of which are high acuity complaints that are clinically important to flag early in triage.